

Research Interests

Multi-modal Learning, Video Understanding, Audio-Language Understanding, Audio-Visual Speech Processing

My research focuses on **multi-modal learning** for deep understanding across audio, visual, and language modalities. Currently, as a postdoctoral researcher at **VIPLab** advised by Prof. Joonseok Lee, I am expanding my prior work on **audio-language understanding** toward broader challenges in **video understanding** and **long-form video summarization**, while also exploring advanced **audio-visual speech synthesis**. My research aims to address fundamental challenges in imbalanced multi-modal data and develop effective cross-modal representations for both comprehensive understanding and controllable generation.

Education

Seoul National University Seoul, Korea

Ph.D. in Intelligence and Information Sep. 2020 – Aug. 2025

Dissertation: “Robust Audio-Language Retrieval under Imbalanced Data Distributions”

Advisor: Prof. Kyogu Lee

Sookmyung Women’s University Seoul, Korea

M.S. in Computer Science Mar. 2015 – Feb. 2017

Thesis: “A Resource Recommendation Method Based on Dynamic Cluster Analysis for Application Characteristics”

Advisor: Prof. Yoonhee Kim

Sookmyung Women’s University Seoul, Korea

B.S. in Mathematics and Computer Science Mar. 2010 – Feb. 2015

Work Experiences

Seoul National University Seoul, Korea

Postdoctoral Researcher Sep. 2025 – current

Korea Institute of Patent Information Daejeon, Korea

Researcher Aug. 2017 – Aug. 2020

Korea Institute of Patent Information Daejeon, Korea

Intern Apr. 2017 – Aug. 2017

Publications

[International Conferences](#)

(* Equal contribution; † Corresponding author)

Watch Your Speech: Text-aware Video-to-Speech Synthesis with Textual Conditioning

Gunwoo Lee*, Yoori Oh* and Yoseob Han

The British Machine Vision Conference (BMVC), 2026

TripleSumm: Adaptive Triple-Modality Fusion for Video Summarization

Sumin Kim*, Hyemin Jeong*, Mingu Kang*, Yejin Kim, **Yoori Oh**[†] and Joonseok Lee[†]

International Conference on Learning Representations (ICLR), 2026

Sounding Highlights: Dual-Pathway Audio Encoders for Audio-Visual Video Highlight Detection

Seohyun Joo and **Yoori Oh**

IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP), 2026

LipSody: Lip-to-Speech Synthesis with Enhanced Prosody Consistency

Jaejun Lee, **Yoori Oh** and Kyogu Lee

IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP), 2026

Understanding Audio-Text Retrieval Through Singular Value Decomposition

Yoori Oh, Yoseob Han, Joonhyeon Bae, Jaeheon Sim and Kyogu Lee

ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR), 2025

Speaking Without Sound : Multi-speaker Silent Speech Voicing with Facial Inputs Only

Jaejun Lee, **Yoori Oh** and Kyogu Lee

IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP), 2025

Distance Sampling-based Paraphraser Leveraging ChatGPT for Text Data Manipulation

Yoori Oh*, Yoseob Han* and Kyogu Lee

ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR), 2024

Hear Your Face : Face-based voice conversion with F0 estimation

Jaejun Lee, **Yoori Oh**, Injune Hwang and Kyogu Lee

INTERSPEECH, 2024

Semi-supervised Learning for Continuous Emotional Intensity Controllable Speech Synthesis with Disentangled Representations

Yoori Oh, Juheon Lee, Yoseob Han and Kyogu Lee

INTERSPEECH, 2023

Exploring Train and Test-Time Augmentations for Audio-Language Learning

Eungbeom Kim*, Jinhee Kim*, **Yoori Oh**, Kyungsu Kim, Minju Park, Jaeheon Sim, Jinwoo Lee and Kyogu Lee

arxiv, 2022

A Hybrid Cloud Resource Clustering Method Using Analysis of Application Characteristic

Yoori Oh and Yoonhee Kim

IEEE 2nd International Workshops on Foundations and Applications of Self Systems (FAS*W), 2017*

A SLA-based Spark Cluster Scaling Method in Cloud Environment

Yoori Oh, Jieun Choi, Eunjung Song, Moonji Kim and Yoonhee Kim

18th Asia-Pacific Network Operations and Management Symposium (APNOMS), 2016

[International Journals](#)

A Resource Recommendation Method based on Dynamic Cluster Analysis of Application Characteristics

Yoori Oh and Yoonhee Kim

Cluster Computing, 2019

[Patents](#)

QUEUE-BASED CONTRASTIVE LEARNING METHOD FOR AUDIO-TEXT RETRIEVAL AND COMPUTER DEVICE THEREOF

Jaeheon Sim, Kyogu Lee, **Yoori Oh**, Yoseob Han and Joonhyun Bae

KR, application number : 10-2025-0194175, 2025. 12

METHOD AND APPARATUS FOR AUGMENTING AUDIO-TEXT MULTIMODAL DATA IN TRAIN AND TEST TIME

Kyogu Lee, Jinhee Kim, Eungbeom Kim, **Yoori Oh**, Jaeheon Sim, Jinwoo Lee, Kyungsu Kim and Minju Park

PCT, application number : PCT/KR2023/017075, 2023. 10

METHOD AND APPARATUS FOR AUGMENTING AUDIO-TEXT MULTIMODAL DATA IN TRAIN AND TEST TIME

Kyogu Lee, Jinhee Kim, Eungbeom Kim, **Yoori Oh**, Jaeheon Sim, Jinwoo Lee, Kyungsu Kim and Minju Park

KR, application number : 10-2023-0146812, 2023. 10

METHOD AND APPARATUS FOR STRUCTURING PATENT CLAIMS

Youngmin Ahn, Hojin Park, Suhyun Oh, Jonghyun Chae, Bong-Gun Lee, Yoonho Jung, Jaeok Min and **Yoori Oh**

KR, registration number : 10-2181677, 2020. 11

MOBILE TERMINAL CONTROL METHOD FOR PERSONALIZED DIETARY COACHING

Yoonhee Kim, Jieun Choi, **Yoori Oh** and Youngran Jeon

KR, registration number : 10-2016263, 2019. 8

Honors and Scholarships

Frontier Fellowship (Seoul National University)	2026
Gold Reviewer Award (ICML 2026)	2026
Teaching assistance scholarship (Seoul National University)	2024
Brain Korea 21 Plus scholarship (Seoul National University)	2020 - 2023
Bronze Award (12th open source software developer competition)	2018. 11
Best Presentation Award (KIISE, KSC 2016)	2017. 01
Bronze Best Paper Award (IEEE Student Paper Contest)	2016. 12
Grand Prize (2015 Convergence App Contest)	2015. 12
Best Paper Award (KSII 2015)	2015. 10

Teaching Experience

Teaching assistant, Seoul National University	Spring 2024
Course title : From Sound to Meaning (Prof. Kyogu Lee)	

Academic Services

Reviewer

- IEEE Transactions on Audio, Speech and Language Processing (TASLP)
- AAAI Conference on Artificial Intelligence (AAAI)
- ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR)
- International Conference on Machine Learning (ICML)
- Conference on Neural Information Processing Systems (NeurIPS)